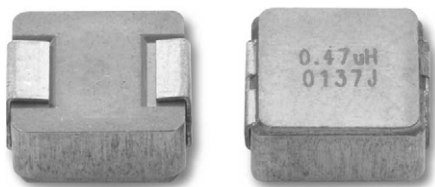




## IHLP® Commercial Inductors, High Saturation Series



### LINKS TO ADDITIONAL RESOURCES



3D Models



Design Tools

### STANDARD ELECTRICAL SPECIFICATIONS

| L <sub>0</sub><br>INDUCTANCE<br>± 20 %<br>AT 100 kHz,<br>0.25 V, 0 A<br>(μH) | DCR<br>TYP.<br>25 °C<br>(mΩ) | DCR<br>MAX.<br>25 °C<br>(mΩ) | HEAT<br>RATING<br>CURRENT<br>DC TYP.<br>(A) <sup>(1)</sup> | SATURATION<br>CURRENT<br>DC TYP.<br>(A) <sup>(2)</sup> | SRF<br>TYP.<br>(MHz) |
|------------------------------------------------------------------------------|------------------------------|------------------------------|------------------------------------------------------------|--------------------------------------------------------|----------------------|
| 0.10                                                                         | 1.5                          | 1.7                          | 32.5                                                       | 60                                                     | 400                  |
| 0.15                                                                         | 1.9                          | 2.5                          | 26                                                         | 52                                                     | 180                  |
| 0.20                                                                         | 2.4                          | 3.0                          | 24                                                         | 41                                                     | 150                  |
| 0.22                                                                         | 2.5                          | 2.8                          | 23                                                         | 40                                                     | 126                  |
| 0.33                                                                         | 3.5                          | 3.9                          | 20                                                         | 30                                                     | 100                  |
| 0.47                                                                         | 4                            | 4.2                          | 17.5                                                       | 26                                                     | 75                   |
| 0.68                                                                         | 5                            | 5.5                          | 15.5                                                       | 25                                                     | 62                   |
| 0.82                                                                         | 6.7                          | 8                            | 13                                                         | 24                                                     | 60                   |
| 1.0                                                                          | 9                            | 10                           | 11                                                         | 22                                                     | 55                   |
| 1.5                                                                          | 14                           | 15                           | 9                                                          | 18                                                     | 40                   |
| 2.2                                                                          | 18                           | 20                           | 8                                                          | 14                                                     | 38                   |
| 3.3                                                                          | 28                           | 30                           | 6                                                          | 13.5                                                   | 30                   |
| 4.7                                                                          | 37                           | 40                           | 5.5                                                        | 10                                                     | 25                   |
| 6.8                                                                          | 54                           | 60                           | 4.5                                                        | 8                                                      | 21                   |
| 8.2                                                                          | 64                           | 68                           | 4                                                          | 7.5                                                    | 17                   |
| 10                                                                           | 102                          | 105                          | 3                                                          | 7.0                                                    | 16                   |

#### Notes

- All test data is referenced to 25 °C ambient
- Operating temperature range -55 °C to +125 °C
- The part temperature (ambient + temp. rise) should not exceed 125 °C under worst case operating conditions. Circuit design, component placement, PWB trace size and thickness, airflow and other cooling provisions all affect the part temperature. Part temperature should be verified in the end application.
- Rated operating voltage (across inductor) = 75 V
- (1) DC current (A) that will cause an approximate ΔT of 40 °C
- (2) DC current (A) that will cause L<sub>0</sub> to drop approximately 20 %

### FEATURES

- Lowest height (3.0 mm) in this package footprint
- Shielded construction
- Excellent DC/DC energy storage up to 5 MHz. Filter inductor applications up to SRF (see "Standard Electrical Specifications" table)
- Lowest DCR/μH, in this package size
- Handles high transient current spikes without saturation
- Ultra low buzz noise, due to composite construction
- IHLP design; PATENT(S): [www.vishay.com/patents](http://www.vishay.com/patents)
- Material categorization: for definitions of compliance please see [www.vishay.com/doc?99912](http://www.vishay.com/doc?99912)

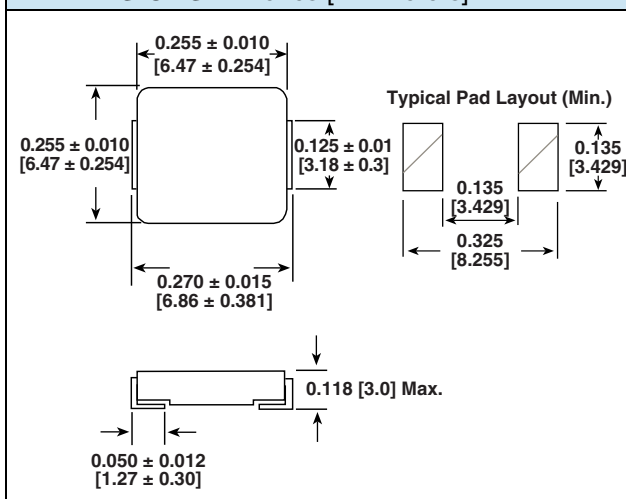


**RoHS**  
COMPLIANT  
HALOGEN  
**FREE**  
**GREEN**  
(5-2008)

### APPLICATIONS

- PDA / notebook / desktop / server applications
- High current POL converters
- Low profile, high current power supplies
- Battery powered devices
- DC/DC converters in distributed power systems
- DC/DC converter for field programmable gate array (FPGA)

### DIMENSIONS in inches [millimeters]



### DESCRIPTION

| IHLP-2525CZ-01 | 1.0 μH           | ± 20 %               | EK           | e3                             |
|----------------|------------------|----------------------|--------------|--------------------------------|
| MODEL          | INDUCTANCE VALUE | INDUCTANCE TOLERANCE | PACKAGE CODE | JEDEC® LEAD (Pb)-FREE STANDARD |

### GLOBAL PART NUMBER

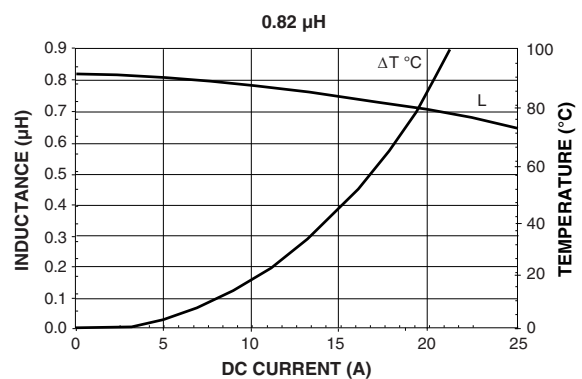
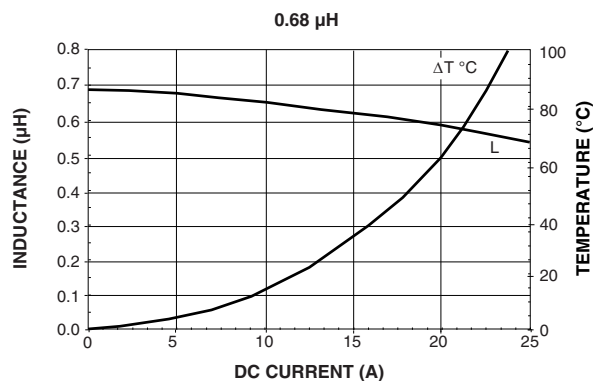
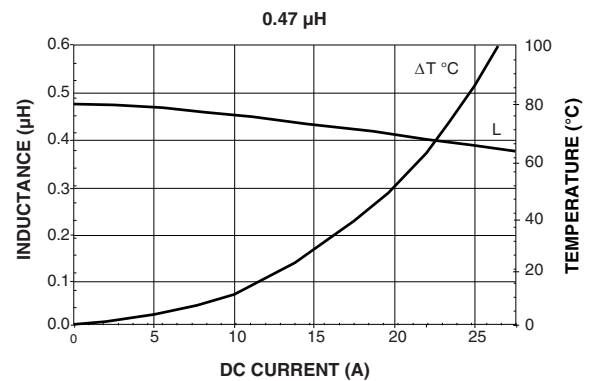
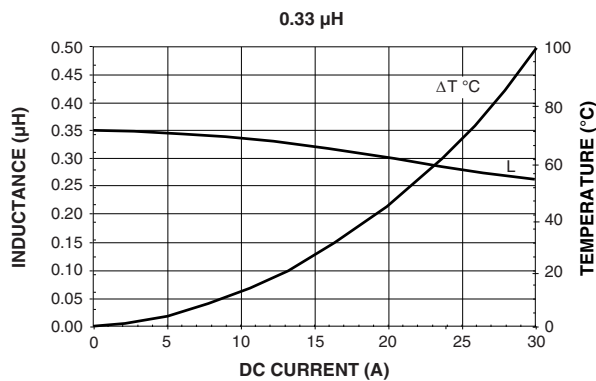
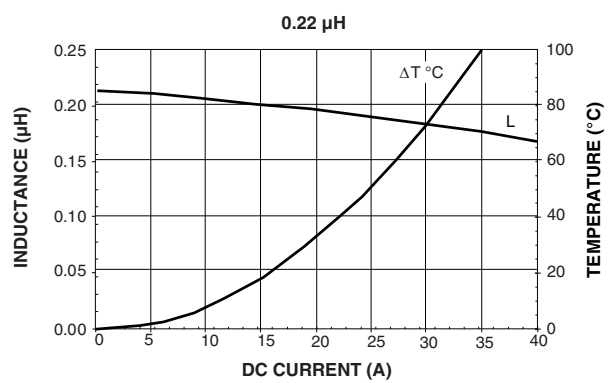
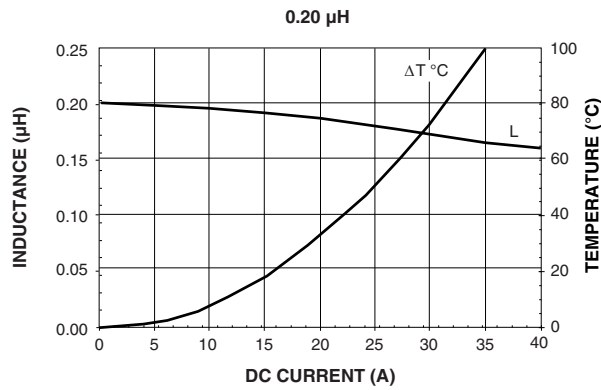
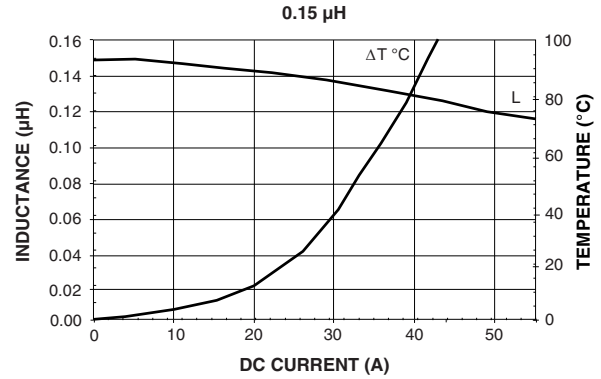
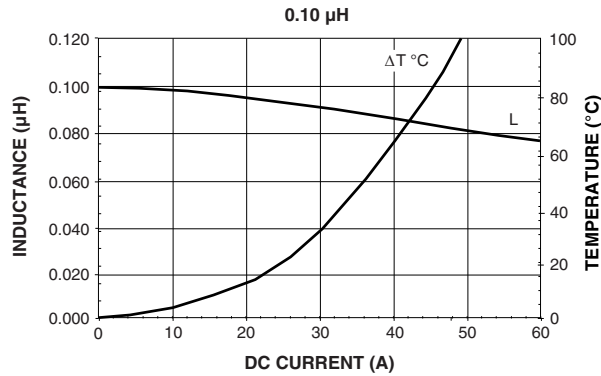
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| I              | H | L | P | 2    | 5 | 2 | 5 | C | Z | E            | K | 1                | R | 0    | M      | 0 | 1 |
| PRODUCT FAMILY |   |   |   | SIZE |   |   |   |   |   | PACKAGE CODE |   | INDUCTANCE VALUE |   | TOL. | SERIES |   |   |

PATENT(S): [www.vishay.com/patents](http://www.vishay.com/patents)

This Vishay product is protected by one or more United States and international patents.

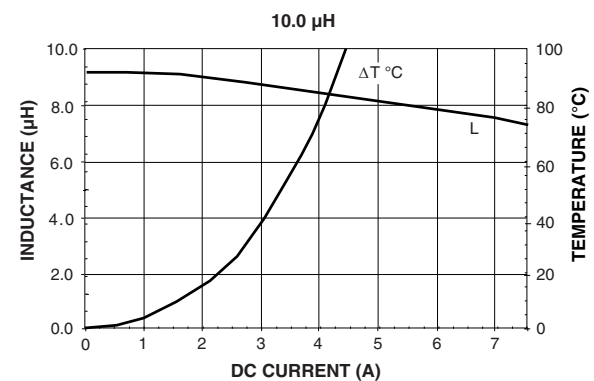
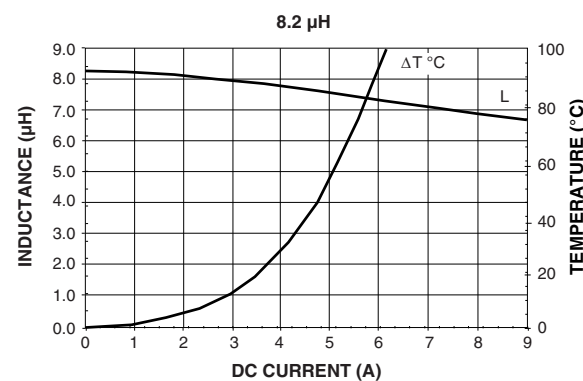
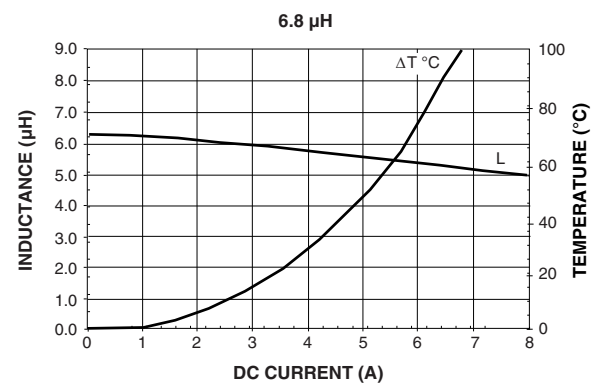
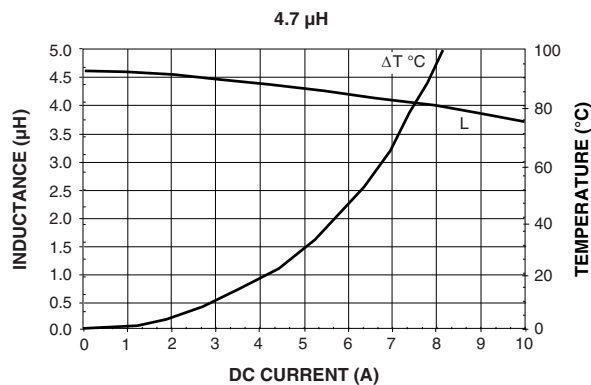
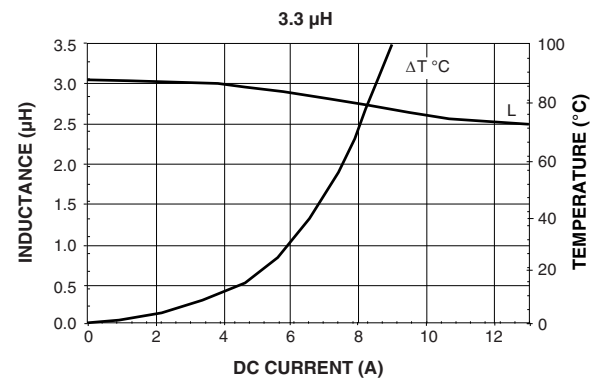
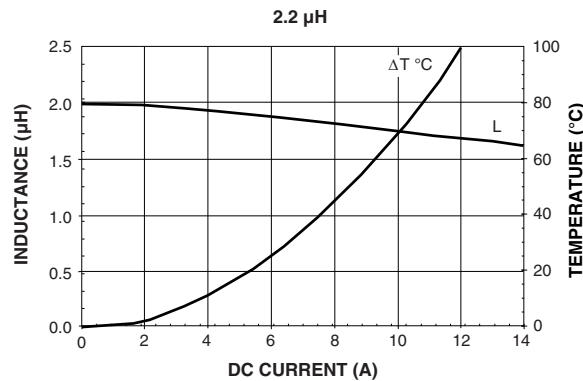
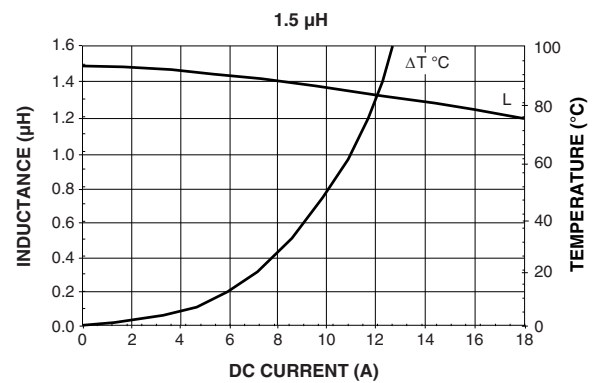
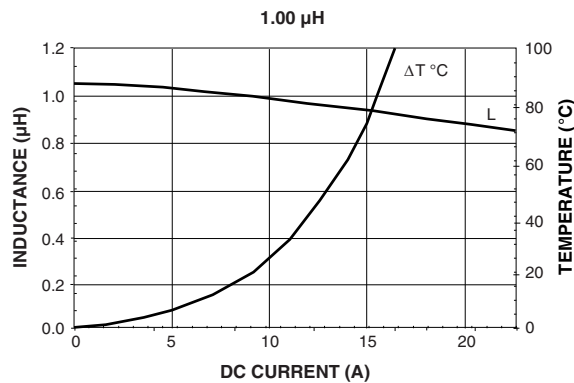


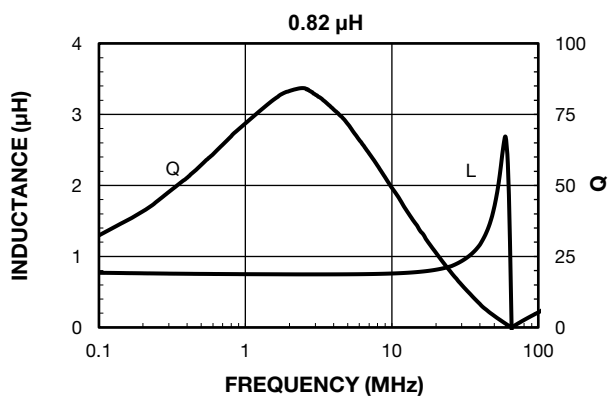
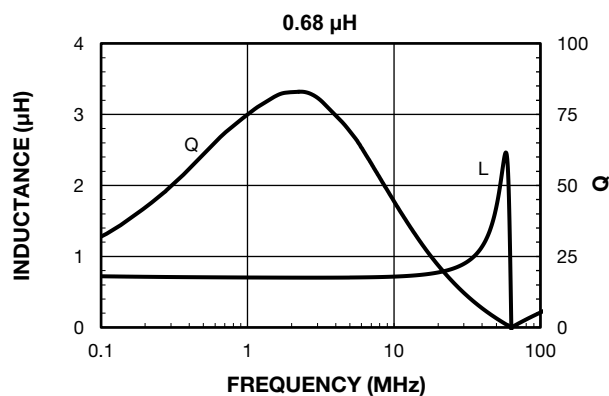
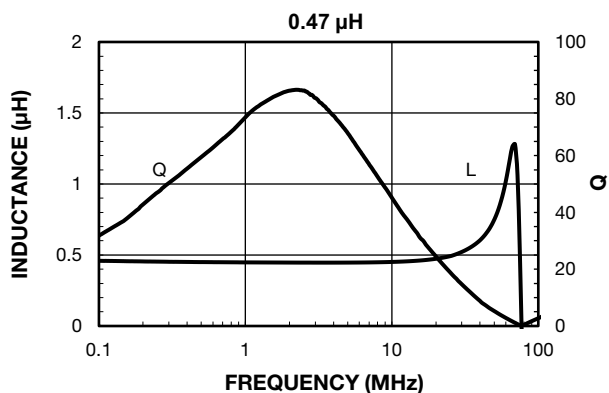
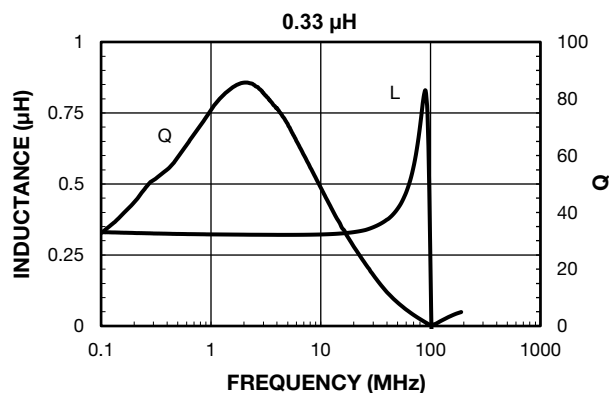
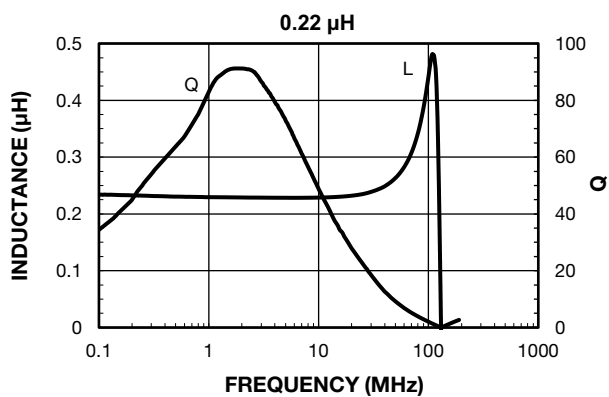
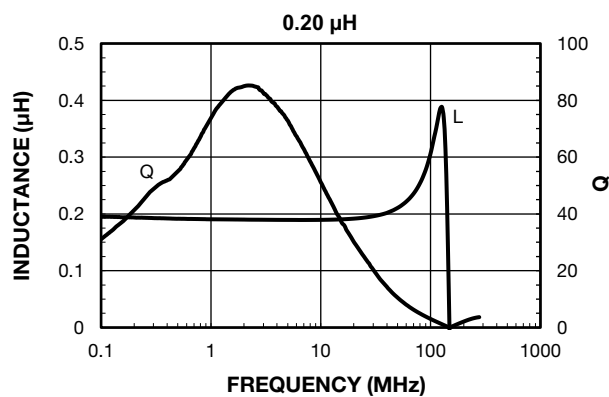
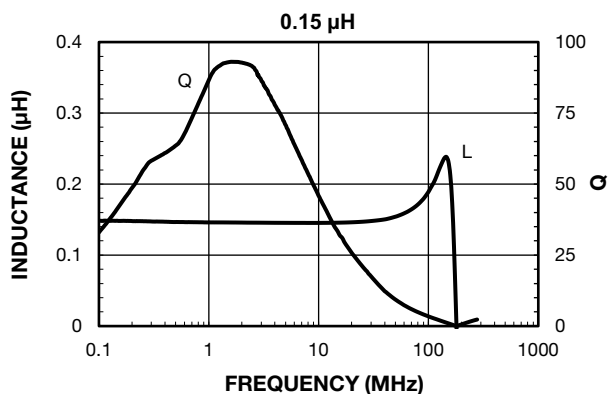
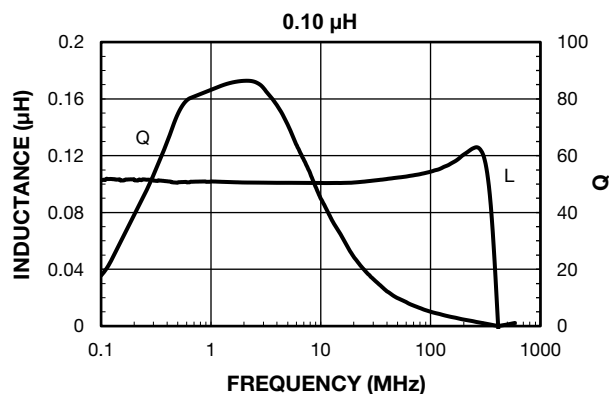
| <b>PACKAGE CODE OPTIONS</b>                             |
|---------------------------------------------------------|
| EK = tape and reel packaging (2500 pcs on 13-inch reel) |
| ER = tape and reel packaging (2000 pcs on 13-inch reel) |

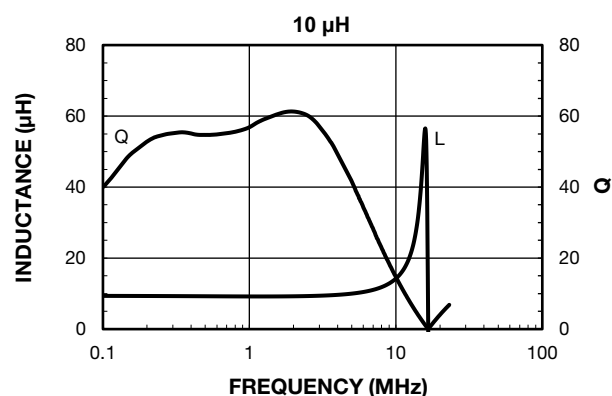
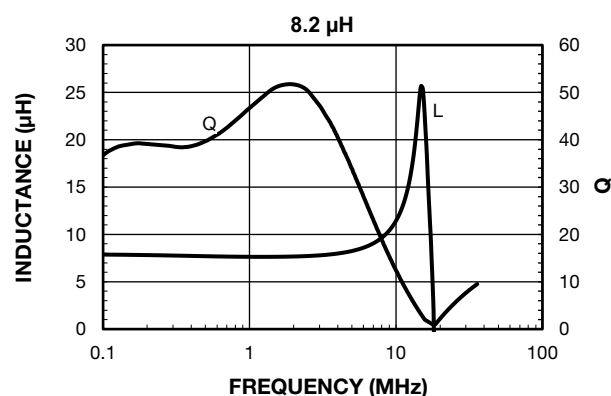
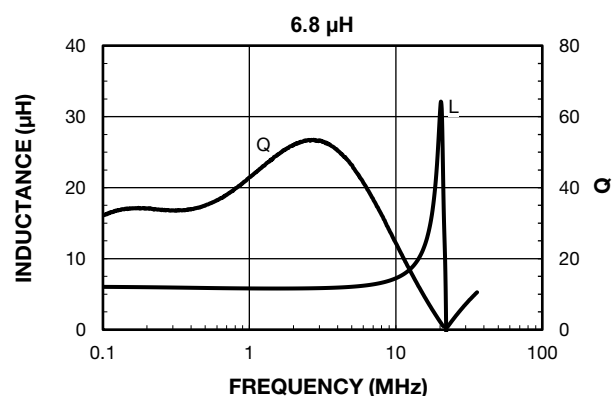
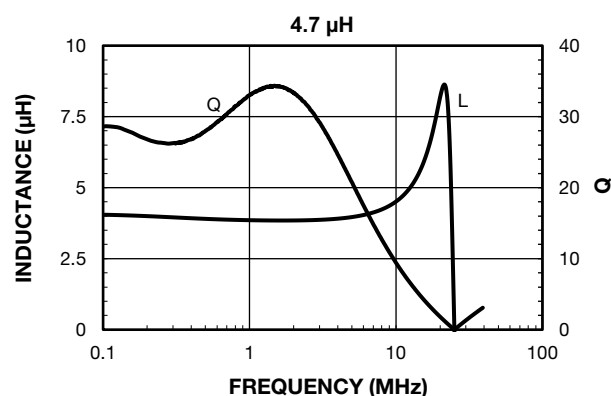
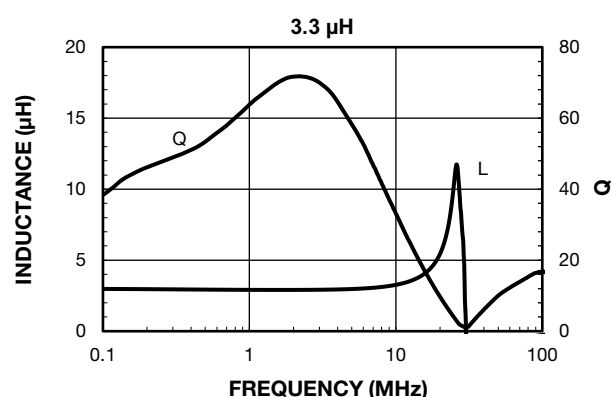
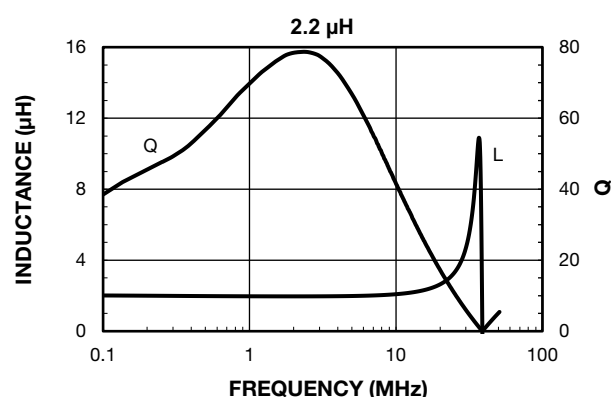
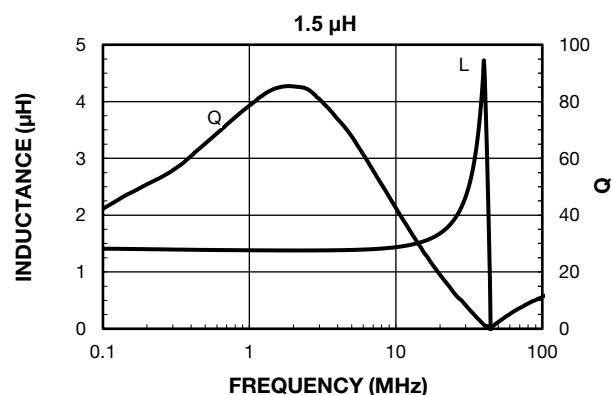
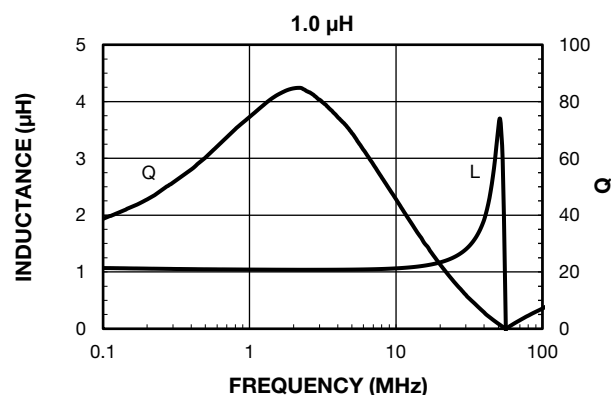
**PERFORMANCE GRAPHS**



## PERFORMANCE GRAPHS



**PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY**

**PERFORMANCE GRAPHS: INDUCTANCE AND Q VS. FREQUENCY**



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